

Atividade_00 – Conceitos básicos. ATmega328 e introdução ao ESP32

1. Baixe os datasheets (folha de dados) do Atmel ATmega32u4, do PIC PIC16F87XA (PIC16F877A), do Texas MSP430F249 e do STMicroelectronics STM8TL53F4.

Compare os quatro em termos de: memória (flash, eeprom, pilha, organização), conjunto de instruções, interrupções, portas de entrada e saída (quantidade, características?), periféricos (timers, ADC, protocolos de comunicação nativos).

DATASHEETS:

1. https://ww1.microchip.com/downloads/en/devicedoc/atmel-7766-8-bit-avr-atmega16u4-32u4_datasheet.pdf
2. <https://www.alldatasheet.com/datasheet-pdf/view/74972/MICROCHIP/PIC16F87X.html>
3. <https://www.alldatasheet.com/datasheet-pdf/view/867564/TI1/MSP430F249.html>
4. <https://www.st.com/resource/en/datasheet/stm8tl52f4.pdf>

Memória:

Modelo	Flash	RAM	EEPROM
ATMega32u4	32 KB	2 KB	1 KB
PIC16F87XA (PIC16F877A)	8 KB x 14 palavras	368 KB	256 KB
MSP430F249	60 KB	2 KB	– Sem menções de EEPROM no datasheet –
STM8TL53F4	16 KB (incluindo os 2 KB EEPROM)	4 KB	2 KB

Conjunto de Instruções:

Modelo	Conjunto de Instruções
ATMega32u4	131 instruções
PIC16F87XA (PIC16F877A)	35 instruções
MSP430F249	Arquitetura RISC de 16 bits, tempo de ciclo de instrução de 62,5 ns
STM8TL53F4	80 instruções

Interrupções:

- *ATMega32u4*:

Interrupt Vectors in *ATmega16U4/ATmega32U4*

Table 9-1. Reset and Interrupt Vectors(cont'd)

Vector No.	Program Address ⁽²⁾	Source	Interrupt Definition
1	\$0000 ⁽¹⁾	RESET	External Pin, Power-on Reset, Brown-out Reset, Watchdog Reset, and JTAG AVR Reset
2	\$0002	INT0	External Interrupt Request 0
3	\$0004	INT1	External Interrupt Request 1
4	\$0006	INT2	External Interrupt Request 2
5	\$0008	INT3	External Interrupt Request 3
6	\$000A	Reserved	Reserved
7	\$000C	Reserved	Reserved
8	\$000E	INT6	External Interrupt Request 6
9	\$0010	Reserved	Reserved
10	\$0012	PCINT0	Pin Change Interrupt Request 0
11	\$0014	USB General	USB General Interrupt request
12	\$0016	USB Endpoint	USB Endpoint Interrupt request
13	\$0018	WDT	Watchdog Time-out Interrupt
14	\$001A	Reserved	Reserved
15	\$001C	Reserved	Reserved
16	\$001E	Reserved	Reserved
17	\$0020	TIMER1 CAPT	Timer/Counter1 Capture Event
18	\$0022	TIMER1 COMPA	Timer/Counter1 Compare Match A
19	\$0024	TIMER1 COMPB	Timer/Counter1 Compare Match B
20	\$0026	TIMER1 COMPC	Timer/Counter1 Compare Match C
21	\$0028	TIMER1 OVF	Timer/Counter1 Overflow
22	\$002A	TIMER0 COMPA	Timer/Counter0 Compare Match A
23	\$002C	TIMER0 COMPB	Timer/Counter0 Compare match B
24	\$002E	TIMER0 OVF	Timer/Counter0 Overflow
25	\$0030	SPI (STC)	SPI Serial Transfer Complete
26	\$0032	USART1 RX	USART1 Rx Complete
27	\$0034	USART1 UDRE	USART1 Data Register Empty

Table 9-1. Reset and Interrupt Vectors(cont'd)

Vector No.	Program Address ⁽²⁾	Source	Interrupt Definition
28	\$0036	USART1TX	USART1 Tx Complete
29	\$0038	ANALOG COMP	Analog Comparator
30	\$003A	ADC	ADC Conversion Complete
31	\$003C	EE READY	EEPROM Ready
32	\$003E	TIMER3 CAPT	Timer/Counter3 Capture Event
33	\$0040	TIMER3 COMPA	Timer/Counter3 Compare Match A
34	\$0042	TIMER3 COMPB	Timer/Counter3 Compare Match B
35	\$0044	TIMER3 COMPC	Timer/Counter3 Compare Match C
36	\$0046	TIMER3 OVF	Timer/Counter3 Overflow
37	\$0048	TWI	2-wire Serial Interface
38	\$004A	SPM READY	Store Program Memory Ready
39	\$004C	TIMER4 COMPA	Timer/Counter4 Compare Match A
40	\$004E	TIMER4 COMPB	Timer/Counter4 Compare Match B
41	\$0050	TIMER4 COMPD	Timer/Counter4 Compare Match D
42	\$0052	TIMER4 OVF	Timer/Counter4 Overflow
43	\$0054	TIMER4 FPF	Timer/Counter4 Fault Protection Interrupt

- Notes:
1. When the BOOTRST Fuse is programmed, the device will jump to the Boot Loader address at reset, see ["Memory Programming" on page 353](#).
 2. When the IVSEL bit in MCUCR is set, Interrupt Vectors will be moved to the start of the Boot Flash Section. The address of each Interrupt Vector will then be the address in this table added to the start address of the Boot Flash Section.

The table shows reset and Interrupt Vectors placement for the various combinations of BOOTRST and IVSEL settings. If the program never enables an interrupt source, the Interrupt Vectors are not used, and regular program code can be placed at these locations. This is also the case if the Reset Vector is in the Application section while the Interrupt Vectors are in the Boot section or vice versa.

Table 9-2. Reset and Interrupt Vectors Placement

BOOTRST	IVSEL	Reset Address	Interrupt Vectors Start Address
1	0	0x0000	0x0002
1	1	0x0000	Boot Reset Address + 0x0002
0	0	Boot Reset Address	0x0002
0	1	Boot Reset Address	Boot Reset Address + 0x0002

- Note:
1. The Boot Reset Address is shown in [Table 27-4 on page 340](#). For the BOOTRST Fuse "1" means unprogrammed while "0" means programmed.

- **PIC16F87XA:**

REGISTER 2-3: INTCON REGISTER (ADDRESS 0Bh, 8Bh, 10Bh, 18Bh)

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-x
GIE	PEIE	TOIE	INTE	RBIE	TOIF	INTF	RBIF

bit 7

bit 0

- bit 7 **GIE:** Global Interrupt Enable bit
1 = Enables all unmasked interrupts
0 = Disables all interrupts
- bit 6 **PEIE:** Peripheral Interrupt Enable bit
1 = Enables all unmasked peripheral interrupts
0 = Disables all peripheral interrupts
- bit 5 **TOIE:** TMR0 Overflow Interrupt Enable bit
1 = Enables the TMR0 interrupt
0 = Disables the TMR0 interrupt
- bit 4 **INTE:** RB0/INT External Interrupt Enable bit
1 = Enables the RB0/INT external interrupt
0 = Disables the RB0/INT external interrupt
- bit 3 **RBIE:** RB Port Change Interrupt Enable bit
1 = Enables the RB port change interrupt
0 = Disables the RB port change interrupt
- bit 2 **TOIF:** TMR0 Overflow Interrupt Flag bit
1 = TMR0 register has overflowed (must be cleared in software)
0 = TMR0 register did not overflow
- bit 1 **INTF:** RB0/INT External Interrupt Flag bit
1 = The RB0/INT external interrupt occurred (must be cleared in software)
0 = The RB0/INT external interrupt did not occur
- bit 0 **RBIF:** RB Port Change Interrupt Flag bit
1 = At least one of the RB7:RB4 pins changed state; a mismatch condition will continue to set the bit. Reading PORTB will end the mismatch condition and allow the bit to be cleared (must be cleared in software).
0 = None of the RB7:RB4 pins have changed state

REGISTER 2-4: PIE1 REGISTER (ADDRESS 8Ch)

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
PSPIE ⁽¹⁾	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE
bit 7				bit 0			

- bit 7 **PSPIE⁽¹⁾**: Parallel Slave Port Read/Write Interrupt Enable bit
 1 = Enables the PSP read/write interrupt
 0 = Disables the PSP read/write interrupt
- bit 6 **ADIE**: A/D Converter Interrupt Enable bit
 1 = Enables the A/D converter interrupt
 0 = Disables the A/D converter interrupt
- bit 5 **RCIE**: USART Receive Interrupt Enable bit
 1 = Enables the USART receive interrupt
 0 = Disables the USART receive interrupt
- bit 4 **TXIE**: USART Transmit Interrupt Enable bit
 1 = Enables the USART transmit interrupt
 0 = Disables the USART transmit interrupt
- bit 3 **SSPIE**: Synchronous Serial Port Interrupt Enable bit
 1 = Enables the SSP interrupt
 0 = Disables the SSP interrupt
- bit 2 **CCP1IE**: CCP1 Interrupt Enable bit
 1 = Enables the CCP1 interrupt
 0 = Disables the CCP1 interrupt
- bit 1 **TMR2IE**: TMR2 to PR2 Match Interrupt Enable bit
 1 = Enables the TMR2 to PR2 match interrupt
 0 = Disables the TMR2 to PR2 match interrupt
- bit 0 **TMR1IE**: TMR1 Overflow Interrupt Enable bit
 1 = Enables the TMR1 overflow interrupt
 0 = Disables the TMR1 overflow interrupt

interrupt, and the EEPROM write operation interrupt.

REGISTER 2-6: PIE2 REGISTER (ADDRESS 8Dh)

U-0	R/W-0	U-0	R/W-0	R/W-0	U-0	U-0	R/W-0
—	Reserved	—	EEIE	BCLIE	—	—	CCP2IE
bit 7				bit 0			

- bit 7 **Unimplemented**: Read as '0'
- bit 6 **Reserved**: Always maintain this bit clear
- bit 5 **Unimplemented**: Read as '0'
- bit 4 **EEIE**: EEPROM Write Operation Interrupt Enable
 1 = Enable EE Write Interrupt
 0 = Disable EE Write Interrupt
- bit 3 **BCLIE**: Bus Collision Interrupt Enable
 1 = Enable Bus Collision Interrupt
 0 = Disable Bus Collision Interrupt
- bit 2-1 **Unimplemented**: Read as '0'
- bit 0 **CCP2IE**: CCP2 Interrupt Enable bit
 1 = Enables the CCP2 interrupt
 0 = Disables the CCP2 interrupt

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

- n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

REGISTER 2-7: PIR2 REGISTER (ADDRESS 0Dh)

U-0	R/W-0	U-0	R/W-0	R/W-0	U-0	U-0	R/W-0
—	Reserved	—	EEIF	BCLIF	—	—	CCP2IF
bit 7				bit 0			

- bit 7 **Unimplemented:** Read as '0'
- bit 6 **Reserved:** Always maintain this bit clear
- bit 5 **Unimplemented:** Read as '0'
- bit 4 **EEIF:** EEPROM Write Operation Interrupt Flag bit
1 = The write operation completed (must be cleared in software)
0 = The write operation is not complete or has not been started
- bit 3 **BCLIF:** Bus Collision Interrupt Flag bit
1 = A bus collision has occurred in the SSP, when configured for I2C Master mode
0 = No bus collision has occurred
- bit 2-1 **Unimplemented:** Read as '0'
- bit 0 **CCP2IF:** CCP2 Interrupt Flag bit
Capture mode:
1 = A TMR1 register capture occurred (must be cleared in software)
0 = No TMR1 register capture occurred
Compare mode:
1 = A TMR1 register compare match occurred (must be cleared in software)
0 = No TMR1 register compare match occurred
PWM mode:
Unused

Legend:

R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'
- n = Value at POR	'1' = Bit is set	'0' = Bit is cleared x = Bit is unknown

- **MSP430D249:**

- The interrupt vectors and the power-up starting address are located in the address range 0xFFFF to 0xFFC0.
- The vector contains the 16-bit address of the appropriate interrupt-handler instruction sequence. If the reset vector (0xFFFE) contains 0xFFFF (e.g., flash is not programmed) the CPU enters LPM4 after power-up.

interrupt vector addresses				
The interrupt vectors and the power-up starting address are located in the address range 0xFFFF to 0xFFC0. The vector contains the 16-bit address of the appropriate interrupt-handler instruction sequence. If the reset vector (0xFFFE) contains 0xFFFF (e.g., flash is not programmed) the CPU enters LPM4 after power-up.				
INTERRUPT SOURCE	INTERRUPT FLAG	SYSTEM INTERRUPT	WORD ADDRESS	PRIORITY
Power-up External reset Watchdog Flash key violation PC out of range (see Note 1)	PORIFG WDTIFG RSTIFG KEYV (see Note 2)	Reset	0xFFFE	31, highest
NMI Oscillator fault Flash memory access violation	NMIIFG OFIFG ACCVIFG (see Notes 2 and 7)	(Non)maskable (Non)maskable (Non)maskable	0xFFFC	30
Timer_B7 (see Note 3)	TBCCR0 CCIFG (see Note 4)	Maskable	0xFFFA	29
Timer_B7 (see Note 3)	TBCCR1 to TBCCR6 CCIFGs, TBIFG (see Notes 2 and 4)	Maskable	0xFFFB	28
Comparator_A+	CAIFG	Maskable	0xFFFF	27
Watchdog timer+	WDTIFG	Maskable	0xFFFF4	26
Timer_A3	TACCR0 CCIFG (see Note 4)	Maskable	0xFFFF2	25
Timer_A3	TACCR1 CCIFG TACCR2 CCIFG TAIFG (see Note 2 and 4)	Maskable	0xFFFF0	24
USCI_A0/USCI_B0 receive USCI_B0 I2C status	UCA0RXIFG, UCB0RXIFG (see Note 2 and 5)	Maskable	0xFFEE	23
USCI_A0/USCI_B0 transmit USCI_B0 I2C receive / transmit	UCA0TXIFG, UCB0TXIFG (see Note 2 and 6)	Maskable	0xFFEC	22
ADC12 (see Note 8)	ADC12IFG (see Notes 2 and 4)	Maskable	0xFFEA	21
			0xFFE8	20
I/O port P2 (eight flags)	P2IFG.0 to P2IFG.7 (see Notes 2 and 4)	Maskable	0xFFE6	19
I/O port P1 (eight flags)	P1IFG.0 to P1IFG.7 (see Notes 2 and 4)	Maskable	0xFFE4	18
USCI_A1/B1 receive	UCA1RXIFG, UCB1RXIFG (see Note 2)	Maskable	0xFFE2	17
USCI_A1/B1 transmit	UCA1TXIFG, UCB1TXIFG (see Note 2)	Maskable	0xFFE0	16
Reserved (see Notes 9 and 10)	Reserved		0xFFDE to 0xFFC0	15 to 0, lowest

NOTES: 1. A reset is executed if the CPU tries to fetch instructions from within the module register memory address range (0x0000 – 0x01FF) or from within unused address ranges.

- **STM8TL53F4:**

- Nested interrupt controller with software priority control
- Up to 22 external interrupt sources

6 Interrupt vector mapping

Table 9. Interrupt mapping

IRQ No.	Source block	Description	Wakeup from Halt mode	Wakeup from Active-halt mode	Wakeup from Wait (WFI mode)	Wakeup from Wait (WFE mode) ⁽¹⁾	Vector address
	RESET	Reset	Yes	Yes	Yes	Yes	0x00 8000
	TRAP	Software interrupt	-	-	-	-	0x00 8004
0	Reserved						0x00 8008
1	FLASH	FLASH end of programming/ write attempted to protected page interrupt	-	-	Yes	Yes	0x00 800C
2	PXS	End of conversion/First conversion completed	-	Yes ⁽²⁾	Yes	Yes	0x00 8010
3	Reserved						0x00 8011 -0x00 8017
4	AWU	Auto wakeup from Halt	-	Yes	Yes	Yes	0x00 8018
5	Reserved						0x00 801C
6	EXTIB	External interrupt port B	Yes	Yes	Yes	Yes	0x00 8020
7	EXTID	External interrupt port D	Yes	Yes	Yes	Yes	0x00 8024
8	EXTI0	External interrupt 0	Yes	Yes	Yes	Yes	0x00 8028
9	EXTI1	External interrupt 1	Yes	Yes	Yes	Yes	0x00 802C
10	EXTI2	External interrupt 2	Yes	Yes	Yes	Yes	0x00 8030
11	EXTI3	External interrupt 3	Yes	Yes	Yes	Yes	0x00 8034
12	EXTI4	External interrupt 4	Yes	Yes	Yes	Yes	0x00 8038
13	EXTI5	External interrupt 5	Yes	Yes	Yes	Yes	0x00 803C
14	EXTI6	External interrupt 6	Yes	Yes	Yes	Yes	0x00 8040
15	EXTI7	External interrupt 7	Yes	Yes	Yes	Yes	0x00 8044
16	Reserved						0x00 8048
17	Reserved						0x00 804C -0x00 804F
18	Reserved						0x00 8050
19	TIM2	TIM2 update/overflow/ trigger/break interrupt	-	-	Yes	Yes	0x00 8054
20	TIM2	TIM2 capture/compare interrupt	-	-	Yes	Yes	0x00 8058
21	TIM3	TIM3 update/overflow/ trigger/break interrupt	-	-	Yes	Yes	0x00 805C

Table 9. Interrupt mapping (continued)

IRQ No.	Source block	Description	Wakeup from Halt mode	Wakeup from Active-halt mode	Wakeup from Wait (WFI mode)	Wakeup from Wait (WFE mode) ⁽¹⁾	Vector address
22	TIM3	TIM3 capture/compare interrupt	-	-	Yes	Yes	0x00 8060
23-24	Reserved						0x00 8064-0x00 806B
25	TIM4	TIM4 update/overflow/trigger interrupt	-	-	Yes	Yes	0x00 806C
26	SPI	SPI TX buffer empty/RX buffer not empty/error/wakeup interrupt	Yes	Yes	Yes	Yes	0x00 8070
27	USART	USART transmit data register empty/transmission complete interrupt	-	-	Yes	Yes	0x00 8074
28	USART	USART received data ready/overflow error/idle line detected/parity error/global error interrupt	-	-	Yes	Yes	0x00 8078
29	I2C	I2C interrupt ⁽³⁾	Yes	Yes	Yes	Yes	0x00 807C

1. The Low power wait mode is entered when executing a WFE instruction in Low power run mode. In WFE mode, the interrupt is served if it has been previously enabled. After processing the interrupt, the processor goes back to WFE mode. When the interrupt is configured as a wakeup event, the CPU wakes up and resumes processing.
2. ProxSense activated before executing HALT instruction.
3. The device is woken up from Halt or Active-halt mode only when the address received matches the interface address.

Portas de Entrada e Saída:

- **ATMega32u4:**
 - 5 Portas
- **PIC16F87XA:**
 - 6 Portas.
- **MSP430D249:**
 - 6 portas
- **STM8TL53F4:**
 - 22 a 23 portas.

Modelo	Portas de Entrada e Saída
ATMega32u4	5 portas
PIC16F87XA (PIC16F877A)	6 portas
MSP430F249	6 portas
STM8TL53F4	22 a 23 portas

Periféricos:

- **ATMega32u4:**
 - Two 8-bit Timer/Counters with Separate Prescaler and Compare Mode
 - One 16-bit Timer/Counter with Separate Prescaler, Compare Mode, and Capture Mode
 - Real Time Counter with Separate Oscillator
 - Six PWM Channels
 - 8-channel 10-bit ADC in TQFP and QFN/MLF package
 - Temperature Measurement
 - 6-channel 10-bit ADC in PDIP Package
 - Temperature Measurement
 - Two Master/Slave SPI Serial Interface
 - One Programmable Serial USART
 - One Byte-oriented 2-wire Serial Interface (Philips I2C compatible)
 - Programmable Watchdog Timer with Separate On-chip Oscillator
 - One On-chip Analog Comparator
 - Interrupt and Wake-up on Pin Change

- **PIC16F87XA:**

- Timer0: 8-bit timer/counter with 8-bit prescaler
- Timer1: 16-bit timer/counter with prescaler, can be incremented during SLEEP via external crystal/clock
- Timer2: 8-bit timer/counter with 8-bit period register, prescaler and postscaler
- Two Capture, Compare, PWM modules- Capture is 16-bit, max. resolution is 12.5 ns- Compare is 16-bit, max. resolution is 200 ns- PWM max. resolution is 10-bit
- 10-bit multi-channel Analog-to-Digital converter
- Synchronous Serial Port (SSP) with SPI (Master mode) and I2C (Master/Slave)
- Universal Synchronous Asynchronous Receiver Transmitter (USART/SCI) with 9-bit address detection
- Parallel Slave Port (PSP) 8-bits wide, with external RD, WR and CS controls (40/44-pin only)

- **MSP430D249:**

- The MSP430F249 series are microcontroller configurations with two built-in 16-bit timers, a fast 12-bit A/D converter, a comparator, four universal serial communication interface (USCI) modules, and up to 48 I/O pins.
- ADC12: The ADC12 module supports fast, 12-bit analog-to-digital conversions.

- **STM8TL53F4:**

0x00 5000	GPIO and Peripheral registers ⁽¹⁾	
0x00 57FF		
0x00 5800		

Table 22. Peripheral current consumption

Symbol	Parameter	Typ. V _{DD} = 3.0 V	Unit
I _{DD} (TIM2)	TIM2 supply current ⁽¹⁾	9	μA/MHz
I _{DD} (TIM3)	TIM3 supply current ⁽¹⁾	9	
I _{DD} (TIM4)	TIM4 timer supply current ⁽¹⁾	4	
I _{DD} (USART)	USART supply current ⁽²⁾	7	
I _{DD} (SPI)	SPI supply current ⁽²⁾	4	
I _{DD} (I ² C)	I2C supply current ⁽²⁾	4	

Table 23. ProxSense peripheral current consumption⁽¹⁾

Symbol	ProxSense transmitter Tx	ProxSense receiver Rx	Typical	Unit
I _{DD} (PXS)	1	1	0.6	mA
	1	4	1.1	
	1	10	2.3	

1. Data based on characterization, not tested in production

- **Bibliotecas:**

- STM8TL5x standard peripheral library;
- STM8TL5x STMTouch library.